RK808 Developer Guide

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Preface

Overview

This document primarily introduces the various sub-modules of the RK808, covering related concepts, functions, DTS configuration, and analysis and troubleshooting of common issues.

Product Version

Chip Models	Kernel Version
RK808	3.10, 4.4, 4.19

Intended Audience

This document (this guide) is primarily intended for the following engineers:

- Technical Support Engineers
- Software Development Engineers

Revision History

Version	Author	Date	Modification Description
V1.0.0	Zhang Qing	2019-11-25	Initial Version
V1.0.1	Huang Ying	2022-05-30	Format Modification

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1. Basics

1.1 Overview

The RK808 is a high-performance PMIC integrating four high-current DCDC converters, eight LDOs, two switches (SWITCH), one RTC, and adjustable power-on sequencing.

Power rails in the system are generally categorized into two types: DCDC and LDO. The general characteristics of these two power types are as follows (please search for detailed information independently):

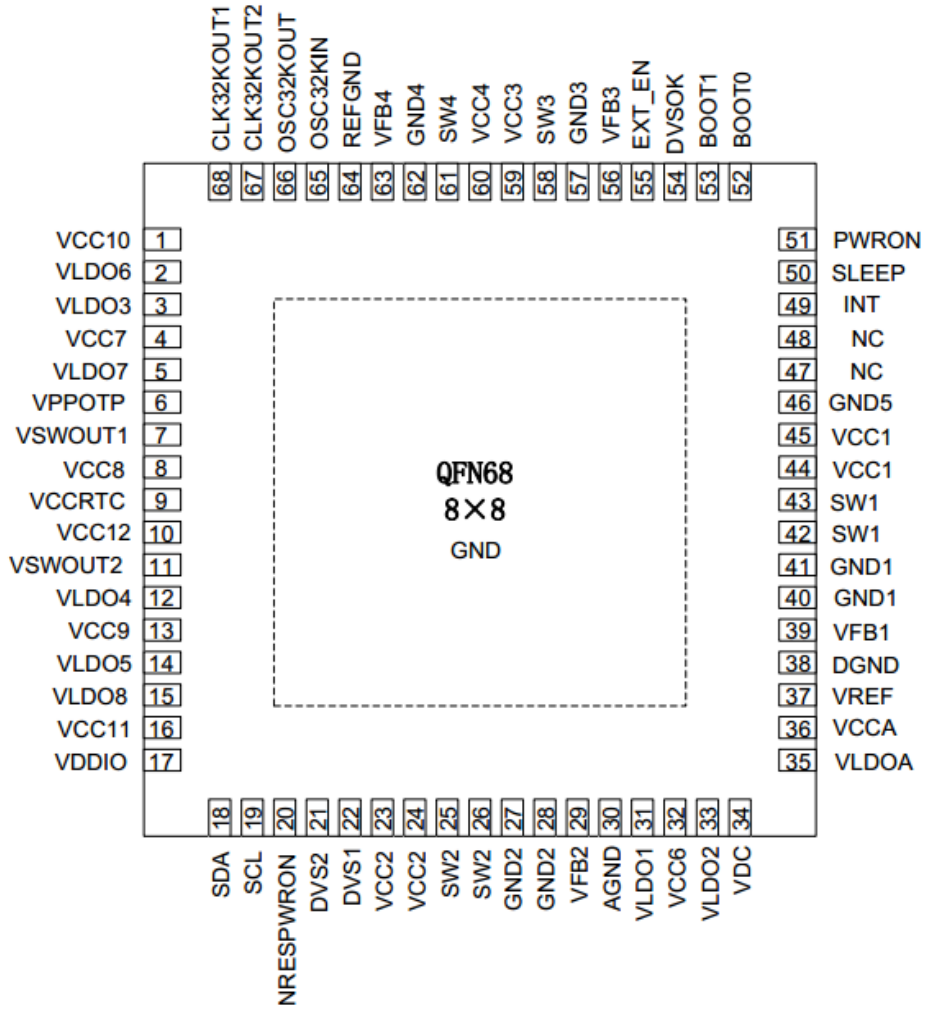
1. DCDC: Offers high efficiency when there is a large difference between input and output voltage, but suffers from relatively large ripple and higher cost. Therefore, it is used for applications with large voltage differences and high current loads. It generally has two operating modes. PWM mode: Good ripple and transient response, but lower efficiency; PFM mode: High efficiency, but poor load capability.
2. LDO: Offers low efficiency and low cost when there is a large difference between input and output voltage. To improve LDO conversion efficiency, the system implements relevant optimizations. For example, if the LDO output voltage is 1.1V, its input voltage can be supplied by the DCDC of VCCIO_3.3V to improve efficiency. Therefore, if the circuit allows, LDOs should be connected to the DCDC output circuit as much as possible, but power-on sequencing must be considered.

1.2 Functions

From the user's perspective, the functions of the RK808 can be summarized into three parts:

1. Regulator function: Controls the status of various DCDC and LDO power rails;
2. RTC function: Provides clock timing, alarm, and other functions;
3. CLK function: Provides two 32.768kHz clock outputs; one is always on and uncontrollable, while the other is software-controllable.

1.3 Chip Pin Functions



In the following description, special attention should be paid to the SLEEP and INT pins:

NO	NAME	SUPPLIES	FUNCTIONAL BLOCK	TYPE	I/O	DESCRIPTION	PU/PD
9	VCCRTC	VCCRTC /AGND	RTC	Power	O	RTC power supply	NO
65	OSC32KIN	VCCRTC /DGND		Analog	I	32KHz crystal oscillator input	NO
66	OSC32KOUT	VCCRTC /DGND		Analog	I	32KHz crystal oscillator output	NO
68	CLK32KOUT1	VCCRTC /DGND		Digital	O	32KHz clock output 1,OD output (always on)	NO
67	CLK32KOUT2	VCCRTC /DGND		Digital	O	32KHz clock output 2,OD output	PD
37	VREF	VCCA /REFGND	REFERENCE	Analog	O	bandgap voltage	PD
64	VREFGND	REFGND		Analog	Gnd	reference ground	NO
36	VCCA	VCCA /GNDA	Analog Power	Power	I	power supply for	NO
6	VPPOTP	VPPOTP /GNDA	Analog Power	Power	I	OTP power supply	NO
45	VCC1	VCC1 /GND1	BUCK1	Power	I/O	buck1 dc-dc power supply	NO
44	VCC1	VCC1 /GND1		Power	I/O	buck1 dc-dc power supply	NO
43	SW1	VCC1 /GND1		Power	I/O	buck1 dc-dc switch output	PD
42	SW1	VCC1 /GND1					
41	GND1	VCC1 /GND1		Power	Gnd	buck1 dc-dc switch ground	NO
40	GND1	VCC1 /GND1					
39	VFB1	VCC1 /REFGND		Analog	I	buck1 dc-dc switch feedback voltage	PD
23	VCC2	VCC2 /GND2	BUCK2	Power	I	buck2 dc-dc power supply	NO

24	VCC2	VCC2 /GND2		Power	I	buck2 dc-dc power supply	NO
25	SW2	VCC2 /GND2		Power	I/O	buck2 dc-dc switch output	PD
26	SW2	VCC2 /GND2		Power	I/O	buck2 dc-dc switch output	PD
27	GND2	VCC2 /GND2		Power	Gnd	buck2 dc-dc switch ground	NO
28	GND2	VCC2 /GND2		Power	Gnd	buck2 dc-dc switch ground	NO
29	VFB2	VCC2 /REFGND		Analog	I	buck2 dc-dc switch feedback voltage	PD
59	VCC3	VCC3 /GND3	BUCK3	Power	I	buck3 dc-dc power supply	NO
58	SW3	VCC3 /GND3		Power	I/O	buck3 dc-dc switch output	PD
57	GND3	VCC3 /GND3		Power	Gnd	buck3 dc-dc switch ground	NO
56	VFB3	VCC3 /REFGND		Analog	I	buck3 dc-dc switch feedback voltage	PD
60	VCC4	VCC4 /GND4	BUCK4	Power	I	buck4 dc-dc power supply	NO
61	SW4	VCC4 /GND4		Power	I/O	buck4 dc-dc switch output	PD
62	GND4	VCC4 /GND4		Power	Gnd	buck4 dc-dc switch ground	NO
63	VFB4	VCC4 /REFGND		Analog	I	buck4 dc-dc switch feedback voltage	PD
47	NC						
46	GND5	VCCA /GND5		Power	Gnd	ground	NO
48	NC						
32	VCC6	VCC6 /AGND	LDO 1~8, SWITCH1,2	Power	I	LDO1,LDO2 power supply	NO
4	VCC7	VCC7 /AGND		Power	I	LDO3,LDO7 power supply	NO
8	VCC8	VCC8 /AGND		Power	I	SWITCH1 power supply	NO

13	VCC9	VCC9 /AGND		Power	I	LDO4,LDO5 power supply	NO
1	VCC10	VCC11 /AGND		Power	I	LDO6 power supply	NO
16	VCC11	VCC11 /AGND		Power	I	LDO8 power supply	NO
10	VCC12	VCC12 /AGND		Power	I	SWITCH2 power supply	NO
31	VLDO1	VCC7 /AGND		Power	O	LDO1 regulator output	PD
33	VLDO2	VCC7 /AGND		Power	O	LDO2 regulator output	PD
3	VLDO3	VCC8 /AGND		Power	O	LDO3 regulator output	PD
12	VLDO4	VCC9 /AGND		Power	O	LDO4 regulator output	PD
14	VLDO5	VCC10 /AGND		Power	O	LDO5 regulator output	PD
2	VLDO6	VCC9 /AGND		Power	O	LDO6 regulator output	PD
5	VLDO7	VCC1 1/AGND		Power	O	LDO7 regulator output	PD
15	VLDO8	VCC11 /AGND		Power	O	LDO8 regulator output	PD
7	VSWOUT1	VCC8 /AGND		Power	O	Switch 1 output	PD
11	VSWOUT2	VCC12 /AGND		Power	O	Switch 2 output	PD
30	AGND	POWER PAD	Analog ground	Power	Gnd	Analog ground	NO
35	VLDOA	POWER PAD	LDOA	Power	I/O	supply for internal analog circuit	NO
38	DGND	POWER PAD	Digital ground	Power	Gnd	Digital ground	NO
17	VDDIO	VDDIO /DGND		Power	I/O	Digital I/O power supply	NO
50	SLEEP	VDDIO /DGND	IO	Digital	I	Active-Sleep state transition control signal	NO
20	NRESPWRON	VDDIO /DGND		Digital	I/O	Power off reset for AP/ External reset digital core(excludes RTC)	PD in power-off state

49	INT	VDDIO /DGND	IO	Digital	O	Interrupt flag (polarity is I2C programmable, default active high)	Programmable PU/PD
51	PWRON	VCCRTC /DGND		Digital	I	External switch-on control signal(ON button)	NO
18	SDA	VDDIO /DGND		Digital	I/O	I2C data signal	NO
19	SCL	VDDIO /DGND		Digital	I/O	I2C clock signal	NO
52	BOOT0	VCCRTC /DGND	IO	Digital	I	Power-up sequence selection	NO
53	BOOT1	VCCRTC /DGND		Digital	I	Power-up sequence selection	NO
55	EXT_EN	VCCRTC /DGND		Digital	O	Output enable for external BUCK in two-battery-cells application	PD
22	DVS1	VDDIO /DGND		Digital	I	BUCK1 DVS voltage /normal voltage transition control signal(polarity is I2C programmable, default active high)	NO
21	DVS2	VDDIO /DGND		Digital	I	BUCK2 DVS voltage /normal voltage transition control signal(polarity is I2C programmable, default active high)	NO
54	DVSOK	VDDIO /DGND		Digital	O	BUCK1 and BUCK2 power good flag after dynamic voltage setting	PD
34	VDC	VDC /AGND		Digital	I	Adapter voltage detect input	NO

1.4 Key Concepts

- I2C Address

7-bit slave address: 0x1b

- The PMIC has three operating modes

1. PMIC Normal Mode

When the system is running normally, the PMIC operates in normal mode, with the pmic_sleep pin held low.

2. PMIC Sleep Mode

To minimize standby power consumption during system suspend, the PMIC switches to sleep mode. In this state, output voltages for certain rails are typically reduced or outputs are disabled entirely, configurable according to specific product requirements. Pulling the pmic_sleep pin high places the PMIC into sleep mode; when the SoC wakes up, pmic_sleep returns to low, causing the PMIC to exit sleep mode.

3. PMIC Shutdown Mode

During system shutdown, the PMIC must power down the entire system. The AP configures the pmic_sleep function to shutdown mode via an I2C command, then pulls the pmic_sleep pin high to enter the shutdown state.

- pmic_sleep Pin

Normally held low, indicating the PMIC is in normal mode. Pulling this pin high switches the PMIC to either sleep or shutdown mode.

- pmic_int Pin

Normally held high. When an Interrupt occurs, it transitions to low. If the Interrupt remains unhandled, the pin stays low.

- pmic_pwron Pin

The power key function requires hardware connection of the power button to this pin. The Driver uses this pin to detect press and release events.

- Operating Modes of DCDC Rails

DCDC converters support PWM (also known as force PWM) and PFM modes. The PMIC also features a mode that dynamically switches between PWM and PFM, commonly referred to as AUTO mode. The PMIC supports two configurations: fixed PWM mode and AUTO PWM/PFM mode. While AUTO mode offers higher efficiency, it exhibits poorer ripple and transient response. For system stability, the PMIC is configured to PWM mode during normal operation and switches to AUTO PWM/PFM mode when the system enters suspend.

- DCDC3 Voltage Regulation

  DCDC3 is a special power rail whose voltage cannot be modified via registers; it can only be adjusted using external voltage-divider resistors. Therefore, if a voltage change is required, the peripheral hardware must be modified. In Rockchip solutions, this rail is typically used as VCC_DDR.

- Runtime Voltage Adjustment Range for DCDC and LDO

1. Continuous DCDC Voltage Range:

Voltage Range (V)	Step Size (mV)	Specific Voltage Levels (V)
0.7125 ~ 1.45	12.5	0.7125, 0.725, 0.7375, ..., 1.45
1.8 ~ 3.3	100	1.8, 1.9, 2.0, 2.2, ..., 3.3

2. Continuous LDO Voltage Range:

Voltage Range (V)	Step Size (mV)	Specific Voltage Levels (V)
0.8 ~ 3.4	100	0.8, 0.9, 1.0, 1.1, 1.2, ..., 3.4

1.5 Power-On Conditions and Sequence

1. Power-On Conditions

The PMIC can be powered on if any of the following conditions is met:

- The EN signal transitions from low to high level.
- The EN signal remains at high level, and an RTC alarm Interrupt is triggered.
- The EN signal remains at high level, and the PWRON button is pressed.

2. Power-On Sequence

Each SoC platform may have different requirements for the power-on sequence of individual power rails. The current power-on sequences are listed below; please refer to the latest datasheet for details:

9 POWER SEQUENCE

	Power On Sequence	Preset Voltage	Power On Sequence	Preset Voltage	Power On Sequence	Preset Voltage	Power On Sequence	Preset Voltage
Boot1, Boot0	00		01		10		11	
BUCK1	4	1.1V/ON	4	1.2V/ON	4	1.0V/ON	OTP	OTP
BUCK2	5	1.1V/ON	5	1.2V/ON	4	1.0V/ON	OTP	OTP
BUCK3	2	1.2V/ON	2	1.2V/ON	3	1.2VON	OTP	OTP
BUCK4	1	3.0V/ON	1	3.0V/ON	1	3.0V/ON	OTP	OTP
LDO1		3.3V/OFF		3.3V/OFF	1	3.3v/ON	OTP	OTP
LDO2		3.3V/OFF	2	3.3V/ON		3.3V/OFF	OTP	OTP
LDO3	3	1.1V/ON	3	1.2V/ON	2	1.0V/ON	OTP	OTP
LDO4	3	2.5V/ON		2.5V/OFF	2	1.8V/ON	OTP	OTP
LDO5		2.8V/OFF		2.8V/OFF		2.8V/OFF	OTP	OTP
LDO6		1.2V/OFF		1.2V/OFF		1.2V/OFF	OTP	OTP
LDO7		1.8/OFF		1.8V/OFF		1.8V/OFF	OTP	OTP
LDO8		3.3V/OFF		1.8V/OFF		3.3V/OFF	OTP	OTP

SWITCH1	1	3.0V/ON	1	3.0V/ON	5	3.0V/ON	OTP	OTP
SWITCH2		3.0V/OFF		3.0V/OFF		3.0V/OFF	OTP	OTP

2. Configuration

2.1 Driver and menuconfig

2.1.1 3.10 Kernel Configuration

RK808 Driver file:

```
drivers/mfd/rk808.c
drivers/mfd/rk808-irq.c
drivers/rtc/rtc-rk808.c
```

For the RK808 DTS file, refer to:

```
arch/arm/boot/dts/rk808.dtsi
arch/arm/boot/dts/rk3288-evb-android-rk808-edp.dts
```

Macro configuration in menuconfig:

```
CONFIG_MFD_RK808
CONFIG_RTC_RK808
```

2.1.2 4.4 Kernel Configuration

RK808 Driver file:

```
drivers/mfd/rk808.c
drivers/rtc/rtc-rk808.c
drivers/regulator/rk808-regulator.c
drivers/clk/clk-rk808.c
```

For the RK808 DTS file, refer to:

```
arch/arm64/boot/dts/rockchip/rk3399-evb-rev3.dtsi
```

Macro configuration corresponding to menuconfig:

```
CONFIG_MFD_RK808
CONFIG_RTC_RK808
CONFIG_REGULATOR_RK808
CONFIG_COMMON_CLK_RK808
```

2.1.3 4.19 Kernel Configuration

RK808 Driver file:

```
drivers/mfd/rk808.c
drivers/rtc/rtc-rk808.c
drivers/regulator/rk808-regulator.c     // Different from the 4.4 kernel
drivers/clk/clk-rk808.c
```

Macro configuration corresponding to menuconfig:

```
CONFIG_MFD_RK808
CONFIG_RTC_RK808
CONFIG_REGULATOR_RK808
CONFIG_COMMON_CLK_RK808
```

2.2 DTS Configuration

2.2.1 3.10 Kernel DTS Configuration

The DTS configuration includes: I2C bus attachment, main device, regulator, RTC, poweroff, and other sections.

```
&i2c1 {
        rk808: rk808@1b {
                reg = <0x1b>;
                status = "okay";
        };
};

/include/ "rk808.dtsi"
&rk808 {
        gpios = <&gpio0 GPIO_A4 GPIO_ACTIVE_HIGH>,
                        <&gpio0 GPIO_B3 GPIO_ACTIVE_LOW>;
        rk808,system-power-controller;

        rtc {
                status = "disabled";
        };

        regulators {
                rk808_dcdc1_reg: regulator@0 {
                        regulator-always-on;
                        regulator-boot-on;
                        regulator-min-microvolt = <750000>;
                        regulator-max-microvolt = <1400000>;
                        regulator-init-microvolt = <1300000>;
                        regulator-name = "vdd_arm";
                        regulator-state-mem {
                                regulator-off-in-suspend;
                        };
```

```
                };
                rk808_dcdc2_reg: regulator@1 {
                                        ..............
                };
                rk808_dcdc3_reg: regulator@2 {
                                        ..............
                };
                ..............................
        };
};
```

1. I2C Attachment

The complete rk808 node must be attached under the corresponding i2c node, with status configured as "okay";

2. Main Section

- Non-modifiable parts

```
rk808,system-power-controller: Declare that the RK808 supports system power-down management.
```

- Modifiable parts

gpios: Specify the pmic_int (first) and pmic_sleep (second) pins;

3. Regulator section

- `regulator-name`: Power supply name; it is recommended to keep it consistent with the hardware schematic, as this name must match when using the regulator_get interface;
- `regulator-min-microvolt`: Minimum voltage adjustable at runtime;
- `regulator-max-microvolt`: Maximum voltage adjustable at runtime;
- `regulator-initial-mode`: DCDC operating mode at runtime, generally configured as 1. 1: force PWM, 2: auto PWM/PFM;
- `regulator-state-mode`: DCDC operating mode during suspend, generally configured as 2. 1: force PWM, 2: auto PWM/PFM;
- `regulator-initial-state`: Mode during suspend; must be configured as 3;
- `regulator-boot-on`: If this property exists, the power supply will be enabled when the regulator is registered;
- `regulator-always-on`: If this property exists, the power supply cannot be disabled at runtime and will be enabled upon registration;
- `regulator-state-enabled`: Maintains power-on state during suspend; to disable this power supply, change it to "regulator-state-disabled";
- `regulator-state-uv`: Standby voltage when power is not cut off during suspend.

Note:

If regulator-min-microvolt and regulator-max-microvolt are set to the same voltage, the system framework will automatically set this voltage and enable the power supply upon regulator registration, requiring no user intervention.

```
If `regulator-boot-on` or `regulator-always-on` is present, the system framework will enable the regulator by default during registration. In this case, the regulator voltage behaves as follows: if `regulator-min-microvolt` and `regulator-max-microvolt` are equal, the system framework sets the voltage to this specified value; if they are unequal, the voltage defaults to the PMIC's hardware default power-on voltage.
```

4. RTC Section

If the RTC function is not required (e.g., on box products), add the node as described above and explicitly set `status = "disabled"`. If enabling is required, either remove the entire RTC node or set its status to `status = "okay"`.

5. Poweroff Section

The RK808 Driver automatically intercepts shutdown commands and executes I2C writes to disable PMIC outputs.

`rk808_shutdown` is registered as a syscore shutdown handler to perform preparatory tasks, such as printing shutdown voltage and disabling RTC Interrupts.

```
static void rk808_shutdown(void)
{
        int ret,i,val;
        u16 reg = 0;
        struct rk808 *rk808 = g_rk808;

        printk("%s\n",__func__);
        /***************get dc1\dc2 voltage ********************/
        for(i=0;i<2;i++){
        reg = rk808_reg_read(rk808,rk808_BUCK_SET_VOL_REG(i));
        reg &= BUCK_VOL_MASK;
        val = 712500 + reg * 12500;
        printk("%s,line=%d dc[%d]= %d\n", __func__,__LINE__,(i+1),val);
        }
        /*************************************************/
        ret = rk808_set_bits(rk808, RK808_INT_STS_MSK_REG1,(0x3<<5),(0x3<<5)); //close rtc
int when power off
        ret = rk808_clear_bits(rk808, RK808_RTC_INT_REG,(0x3<<2)); //close rtc int when power
off
        mutex_lock(&rk808->io_lock);
        mdelay(100);
}

static struct syscore_ops rk808_syscore_ops = {
        .shutdown = rk808_shutdown,
};
```

rk808_device_shutdown actually writes to I2C to disable the PMIC output.

```
static void rk808_device_shutdown(void)
{
        int ret,i;
```

```
        u8 reg = 0;
        struct rk808 *rk808 = g_rk808;
        for(i=0;i < 10;i++){
                printk("%s\n",__func__);
                ret = rk808_i2c_read(rk808,RK808_DEVCTRL_REG,1,&reg);
                if(ret < 0)
                        continue;
                ret = rk808_i2c_write(rk808, RK808_DEVCTRL_REG, 1,(reg |(0x1 <<3)));
                if (ret < 0) {
                        printk("rk808 power off error!\n");
                        continue;
                }
        }
        while(1)wfi();
}
EXPORT_SYMBOL_GPL(rk808_device_shutdown);
```

2.2.2 4.4 Kernel DTS Configuration

DTS configuration includes: I2C bus attachment, main device node, RTC, clock, regulator, and other sections.

```
&i2c1 {
        status = "okay";
        rk808: pmic@1b {
                compatible = "rockchip,rk808";
                reg = <0x1b>;
                interrupt-parent = <&gpio1>;
                interrupts = <21 IRQ_TYPE_LEVEL_LOW>;
                pinctrl-names = "default";
                pinctrl-0 = <&pmic_int_l &pmic_dvs2>;
                rockchip,system-power-controller;
                wakeup-source;
                #clock-cells = <1>;
                clock-output-names = "rk808-clkout1", "rk808-clkout2";

                vcc1-supply = <&vcc3v3_sys>;
                vcc2-supply = <&vcc3v3_sys>;
                vcc3-supply = <&vcc3v3_sys>;
                vcc4-supply = <&vcc3v3_sys>;
                vcc6-supply = <&vcc3v3_sys>;
                vcc7-supply = <&vcc3v3_sys>;
                vcc8-supply = <&vcc3v3_sys>;
                vcc9-supply = <&vcc3v3_sys>;
                vcc10-supply = <&vcc3v3_sys>;
                vcc11-supply = <&vcc3v3_sys>;
                vcc12-supply = <&vcc3v3_sys>;
                vddio-supply = <&vcc1v8_pmu>;

                regulators {
                        vdd_log: DCDC_REG1 {
                                regulator-always-on;
```

```
                                regulator-boot-on;
                                regulator-min-microvolt = <750000>;
                                regulator-max-microvolt = <1350000>;
                                regulator-ramp-delay = <6001>;
                                regulator-name = "vdd_log";
                                regulator-state-mem {
                                        regulator-on-in-suspend;
                                        regulator-suspend-microvolt = <900000>;
                                };
                        };
                        vdd_cpu_l: DCDC_REG2 {
                                regulator-always-on;
                                regulator-boot-on;
                                regulator-min-microvolt = <750000>;
                                regulator-max-microvolt = <1350000>;
                                regulator-ramp-delay = <6001>;
                                regulator-name = "vdd_cpu_l";
                                regulator-state-mem {
                                        regulator-off-in-suspend;
                                };
                        };
                        vcc_ddr: DCDC_REG3 {
                                .................
                        };
                        ..............................
                };
        };
};
```

1. I2C Attachment

The complete RK808 node must be attached under the corresponding I2C node, with `status = "okay";` configured.

2. Main Section

- Non-modifiable parts:

```
compatible = "rockchip,rk808";
reg = <0x1b>;
rockchip,system-power-controller;
wakeup-source;
#clock-cells = <1>;
```

- Modifiable (according to pinctrl rules)

interrupt-parent: Specifies which GPIO the pmic_int belongs to; interrupts: Specifies the pin index and polarity of pmic_int on the GPIO defined in interrupt-parent; pinctrl-names: Do not modify; fixed as "default"; pinctrl-0: References the pmic_int pin defined in pinctrl;

3. rtc

If this module is selected in menuconfig but the drivers do not need to be enabled in practice, you can add an rtc node in the DTS and explicitly set its status to status = "disabled". This prevents the drivers from being enabled, although error logs will appear during boot and can be ignored. To enable the drivers, either remove the corresponding node or set the status to status = "okay".

4. regulator

- `regulator-compatible`: The name used for matching during Driver registration. Do not modify this value, or the Driver will fail to load.
- `regulator-name`: The power supply name. It is recommended to keep this consistent with the hardware schematic, as the `regulator_get` interface requires a match with this name.
- `regulator-init-microvolt`: The initialization voltage during the U-Boot stage; this is invalid in the kernel stage.
- `regulator-min-microvolt`: The minimum voltage adjustable during runtime.
- `regulator-max-microvolt`: The maximum voltage adjustable during runtime.
- `regulator-initial-mode`: The DCDC operating mode during runtime, typically configured as 1. Values: 1: force PWM, 2: auto PWM/PFM.
- `regulator-mode`: The DCDC operating mode during sleep, typically configured as 2. Values: 1: force PWM, 2: auto PWM/PFM.
- `regulator-initial-state`: The mode during suspend; this must be configured as 3.
- `regulator-boot-on`: When this property is present, the power supply is enabled upon Driver registration.
- `regulator-always-on`: When this property is present, it indicates that the power supply cannot be disabled during runtime and will be enabled upon Driver registration.
- `regulator-ramp-delay`: The DCDC voltage ramp-up time, fixed at 12500.
- `regulator-on-in-suspend`: Maintains the power-on state during sleep. To disable this power supply during sleep, change this to `regulator-off-in-suspend`.
- `regulator-suspend-microvolt`: The standby voltage when power is maintained during sleep.

5. Power-off Section

On kernel 4.4, `pm_power_off_prepare` is used to implement pre-shutdown preparations for the PMIC, such as disabling RTC interrupts and configuring specific registers. Register `syscore_shutdown` to perform the actual PMIC shutdown.

6. Clock Section

If a node needs to reference an RK808 clock, use the following format:

`clocks = <&rk808 1>;` First parameter: `&rk808` is fixed and must not be changed. Second parameter: Specifies which RK808 clock to reference; valid values are 0 or 1, where 0 corresponds to `rk808-clkout1` and 1 corresponds to `rk808-clkout2`.

2.2.3 4.19 Kernel DTS Configuration

Refer to the kernel 4.4 DTS configuration. Difference: The kernel 4.19 DTS configuration no longer requires the `gpio` subnode, but other modules still reference and use RK808 pins using the format `gpios = <&rk808 0 GPIO_ACTIVE_LOW>;`.

2.3 Function Interfaces

The following interfaces generally meet daily usage requirements, including regulator enable/disable, voltage setting, and voltage retrieval:

1. Obtain a regulator:

`struct regulator *regulator_get(struct device *dev, const char *id)`

Set `dev` to NULL by default; `id` corresponds to the `regulator-name` property in the DTS.

2. Release a regulator:

`void regulator_put(struct regulator *regulator)`

3. Enable a regulator:

`int regulator_enable(struct regulator *regulator)`

4. Disable a regulator:

`int regulator_disable(struct regulator *regulator)`

5. Obtain regulator voltage

`int regulator_get_voltage(struct regulator *regulator)`

6. Set regulator voltage

`int regulator_set_voltage(struct regulator *regulator, int min_uV, int max_uV)`

The caller must ensure that min_uV equals max_uV.

7. Example

```
struct regulator *rdev_logic;

rdev_logic = regulator_get(NULL, "vdd_logic");                  // Get vdd_logic
regulator_enable(rdev_logic);                                                  // Enable
vdd_logic
regulator_set_voltage(rdev_logic, 1100000, 1100000);    // Set voltage to 1.1V
regulator_disable(rdev_logic);                                                 // Disable
vdd_logic
regulator_put(rdev_logic);                                                            //
Release vdd_logic
```

Note: Kernels 4.4 and 4.19 also provide `devm_`-prefixed regulator interfaces to help developers manage requested resources.

3. Debug

3.1 Kernel 3.10

Since the logic involved in PMIC drivers is not complex, the focus lies primarily on the final register configurations. Therefore, the common debugging method is to directly inspect RK808 registers via the following node:

`/sys/rk808/rk808_test`

Read a register:

`echo r [addr] > /sys/rk808/rk808_test`

Write a register:

`echo w [addr] [value] > /sys/rk808/rk808_test`

It is recommended to read the register again after a write operation to verify success.

3.2 Kernel 4.4

The command format is identical to that of kernel 3.10; only the node path differs. The debug node path for kernel 4.4 is:

`/sys/rk8xx/rk8xx_dbg`

3.3 Kernel 4.19

Refer to the commands for kernel 4.4.